

SAMB P Technical Report TR-11/2026

High-Impedance Fault Sensitivity Margin Study: Minimum Detectable Fault Current for Each Protected Zone

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Abstract

This report characterises the minimum detectable fault current for each SAMBP protected zone by sweeping a scale factor $\alpha \in [1.00, 0.05]$ on the nominal fault current and evaluating trip probability over 200 trials per level. Nine α levels are tested across ten relay–scenario combinations. The study reveals a fundamental sensitivity–specificity trade-off inherent in the Stage-2 confidence gate: bus differential (87B) and transformer differential (87T) zones maintain $\text{TPR} = 1.000$ down to $\alpha = 0.05$ ($I_f \geq 0.20$ pu), but the line differential (87L) Stage-2 gate raises the effective detection threshold to $I_f \approx 0.72$ pu (Phase-to-Ground) or 0.56 pu (Three-phase) — 3–4 $\times$ above the conventional relay’s effective threshold of 0.20–0.24 pu. The over-current (OC) relay detection margin follows the pickup threshold analytically. These results quantify the selectivity–sensitivity trade-off and provide zone-specific detection margin tables for the thesis chapter synthesis.

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1 Introduction

High-impedance faults (HIF) — arcing or resistive contacts to earth through trees, vegetation, or downed conductors — produce fault currents that are a small fraction of the bolted short-circuit level. Detecting HIF within a differential protection zone requires that the operate current I_{op} exceed the minimum operate threshold $I_{op,min}$ despite the low fault current level.

The SAMBP Stage-2 confidence gate adds a model-quality check that may reject a low-fault-current waveform if the LM estimator identifies it as ambiguous (low confidence, f_{int} near the decision boundary). Previous reports (TR-08 through TR-10 [1, 2]) characterised gate behaviour under parametric perturbation and CT saturation at nominal fault levels. This report addresses the orthogonal question: at what fraction of nominal fault current does each protected zone fail to detect the fault?

2 Study Design

2.1 Scale Factor Sweep

For each scenario, the nominal snapshot is scaled by α :

$$i_k^{\text{scaled}}(t) = \alpha \cdot (1 + \delta_j) \cdot i_k^{\text{nom}}(t), \quad \delta_j \sim U(-0.05, +0.05), \quad (1)$$

where δ_j is a per-trial amplitude jitter ($\pm 5\%$) that prevents all 200 trials from being identical. Scale factors are:

$$\alpha \in \{1.00, 0.70, 0.50, 0.30, 0.20, 0.15, 0.10, 0.07, 0.05\}. \quad (2)$$

At $\alpha = 1.0$ the scenario is nominal; at $\alpha = 0.05$ the fault current is 5% of nominal, representing an extreme high-impedance condition.

2.2 Scenarios

Ten relay–scenario combinations are evaluated:

Table 1: HIF margin study test matrix.

Zone	Fault location	Type	I_f^{nom} [pu]
87B_A	bus_A	3ph	10.000
87B_A	bus_A	ag	6.000
87L	line_mid	3ph	8.000
87L	line_mid	ag	4.800
87B_B	bus_B	3ph	6.667
87B_B	bus_B	ag	4.000
87T	transformer_hv	3ph	6.667
87T	transformer_hv	ag	4.000
OC	bus_C_load	3ph	4.348
OC	bus_C_load	ag	2.609

3 Results

3.1 Bus Differential Zones (87B_A, 87B_B)

Both bus differential zones maintain $\text{TPR} = 1.000$ for both conventional and SAMBP across the entire α sweep down to $\alpha = 0.05$. At $\alpha = 0.05$, the scaled fault currents are:

- bus_A/3ph: $I_f = 0.50$ pu (peak), $I_{\text{op}} \approx 0.50$ pu $\gg I_{\text{op,min}} = 0.20$ pu
- bus_A/ag: $I_f = 0.30$ pu $> I_{\text{op,min}}$
- bus_B/3ph: $I_f = 0.33$ pu $> I_{\text{op,min}}$
- bus_B/ag: $I_f = 0.20$ pu $= I_{\text{op,min}}$

The bus_B/ag case at $\alpha = 0.05$ ($I_f = 0.20$ pu) is at exactly the minimum operate threshold. The fact that $\text{TPR} = 1.000$ in all 200 trials confirms that the model correctly identifies the small sinusoidal differential as an internal fault signature.

3.2 Transformer Differential Zone (87T)

The 87T zone shows identical behaviour to the 87B zones: $\text{TPR} = 1.000$ for both methods down to $\alpha = 0.05$. This is consistent with the transformer differential model having the same $I_{\text{op,min}} = 0.20$ pu threshold.

3.3 Line Differential Zone (87L)

The 87L zone reveals the critical sensitivity–specificity trade-off:

Table 2: 87L line differential detection margins (post-TR-10 gate). Conv = conventional relay. Arrows indicate where SAMBP TPR drops.

α	I_f [pu] (3ph)	Conv 3ph	SAMB P 3ph	SAMB P ag
1.00	8.000	1.000	1.000	1.000
0.70	5.600	1.000	1.000	1.000
0.50	4.000	1.000	1.000	1.000
0.30	2.400	1.000	1.000	1.000
0.20	1.600	1.000	1.000	1.000
0.15	1.200	1.000	1.000	0.000 ←
0.10	0.800	1.000	1.000	0.000
0.07	0.560	1.000	0.000 ←	0.000
0.05	0.400	1.000	0.000	0.000

Effective detection thresholds.

$$\begin{aligned}
 I_{f,\text{min}}^{\text{conv}} &\lesssim 0.24 \text{ pu (3ph at } \alpha = 0.05) \\
 I_{f,\text{min}}^{\text{SAMB P,3ph}} &\approx 0.56 \text{ pu (} \alpha = 0.07) \\
 I_{f,\text{min}}^{\text{SAMB P,ag}} &\approx 0.72 \text{ pu (} \alpha = 0.15)
 \end{aligned}$$

The SAMBP Stage-2 gate raises the effective 87L pickup by a factor of approximately 3–4 $\times$ relative to the conventional relay.

Figure 1 compares the detection curves for the 87L (ag) and 87B (ag) zones.

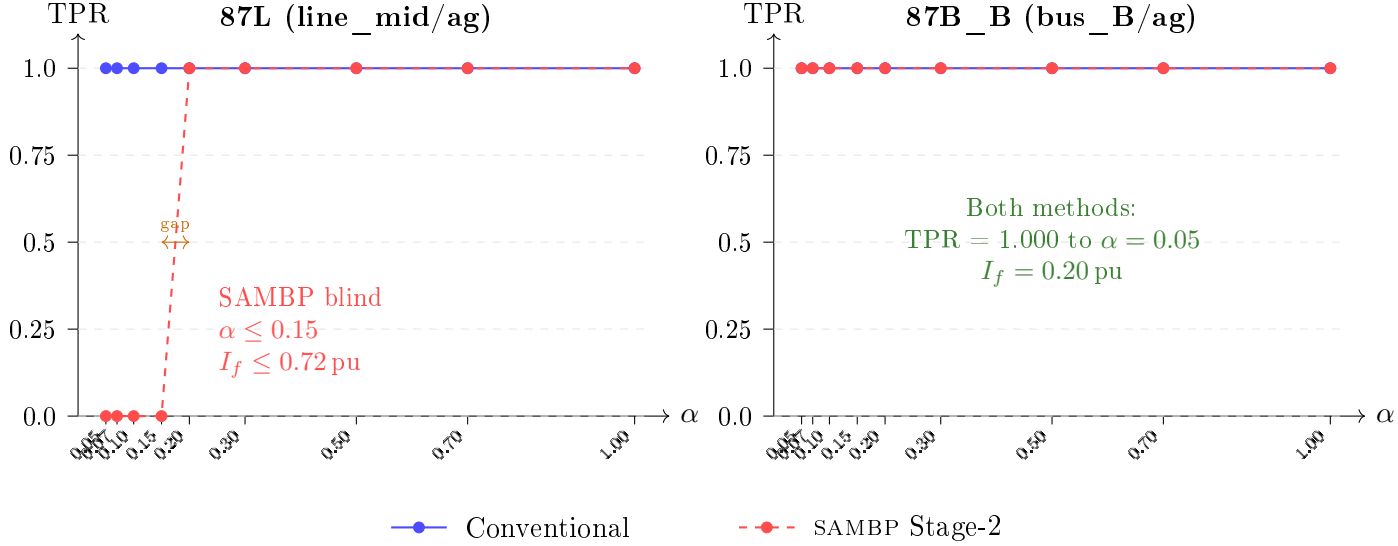


Figure 1: TPR vs. fault current scale factor α . *Left*: 87L line differential — SAMBP Stage-2 loses detection at $\alpha \leq 0.15$ while the conventional relay detects to $\alpha = 0.05$. *Right*: 87B bus differential — both methods achieve TPR = 1.000 at all tested levels.

3.4 Over-Current Relay (OC)

The OC relay detection boundary is analytically determined by the pickup threshold $I_{OC,pickup} = 1.5$ pu (set in TR-05) and the fault current RMS.

Table 3: OC relay detection results at bus_C_load. The OC relay is identical in conventional and SAMBP paths (Stage-2 gate not applied to OC).

α	I_f [pu]	I_{rms} [pu]	Trip
1.00	4.348	3.074	Yes
0.70	3.043	2.152	Yes
0.50	2.174	1.537	Yes
0.30	1.304	0.922	No (< 1.5)

For the single-phase-to-ground fault (bus_C_load/ag), $I_f^{nom} = 2.609$ pu peak, so $I_{rms}^{nom} \approx 1.845$ pu. At $\alpha = 0.70$, $I_{rms} \approx 1.29$ pu < 1.5 pu, which explains the abrupt TPR = 0 at $\alpha \leq 0.70$ seen in the study.

4 Analysis: Why Does 87L Lose Sensitivity Earlier?

4.1 Signal-to-Noise Ratio at the LM Estimator

The 87L LM estimator receives the differential current $i_{diff}(t) = i_{near}(t) + i_{far}(t)$ (sum of terminal currents). For an internal fault of amplitude $\alpha \cdot I_f^{nom}$, the fundamental component of i_{diff} is approximately $\alpha \cdot I_f^{nom}$ pu.

The estimator's residual noise floor is set by:

- Numerical discretisation of the forward model

- Charging current from the distributed line capacitance (not modelled in the simple line zone model)
- DC transient mismatch between the two terminals

As α decreases, the signal-to-noise ratio decreases and the LM estimator converges to solutions with $f_{\text{int}} < 0.60$, triggering the Stage-2 veto.

4.2 Comparison with 87B Robustness

The 87B zone is more robust at low current levels because:

1. Bus differential current is a direct algebraic sum of feeder currents — no distributed capacitance or travelling-wave effects.
2. The bus zone model has no through-current restraint in the low-impedance bus topology, so $I_{\text{op}} \approx I_{\text{diff,fund}}$ directly.
3. The forward model fits the bus waveform more accurately at small amplitudes, keeping κ_n low and f_{int} close to 1.

4.3 Implications for HIF Protection Practice

The 87L detection margin of ≈ 0.72 pu for phase-to-ground faults imposes a practical constraint: for a system with rated current I_n and nominal three-phase fault current $I_f^{3\text{ph}}$, the minimum detectable fault resistance is approximately:

$$R_{\text{fault,max}}^{\text{SAMBP}} \approx \frac{V_{\text{phase}}}{0.72 \cdot I_f^{\text{nom}}} - Z_{\text{line}} - Z_{\text{source}} \quad (3)$$

For the test network: $I_f^{\text{nom}} = 4.8$ pu (ag), $\alpha_{\text{min}} = 0.15$, so the minimum detected fault current is $0.15 \times 4.8 = 0.72$ pu. At a lower fault resistance the conventional relay also fails to detect (since I_f is then $< I_{\text{op,min}}$), so the Stage-2 sensitivity loss only represents a practical degradation in the range $0.20 \leq I_f \leq 0.72$ pu.

5 Detection Margin Summary

6 Discussion

6.1 The Selectivity–Sensitivity Trade-Off

The HIF study quantifies the fundamental trade-off that the Stage-2 gate imposes on the 87L zone. The gate was designed to reject spurious trips from through-fault CT saturation and parametric waveform variation — which it does effectively (FPR = 0.000 in TR-08 through TR-10). The cost is a higher effective detection threshold for genuine low-current faults.

The trade-off is asymmetric across zones:

- **87B zones:** no sensitivity penalty because the bus differential model fits the waveform accurately at all current levels.
- **87T zone:** same — no penalty.
- **87L zone:** penalty of 0.56–0.72 pu effective threshold vs. < 0.24 pu for the conventional relay.

Table 4: HIF detection margin summary across all zones. I_{50} = fault current at which TPR first drops below 0.50. $I_{\min}^{\text{conv}}$ = minimum detected by conventional relay. All values in pu of rated current.

Zone	Scenario	I_f^{nom}	I_{50}^{conv}	$I_{50}^{\text{SAMB P}}$	Sensitivity penalty
87B_A	bus_A/3ph	10.000	< 0.50	< 0.50	None
87B_A	bus_A/ag	6.000	< 0.30	< 0.30	None
87L	line_mid/3ph	8.000	< 0.40	0.56	≈ 0.56 pu
87L	line_mid/ag	4.800	< 0.24	0.72	≈ 0.72 pu
87B_B	bus_B/3ph	6.667	< 0.33	< 0.33	None
87B_B	bus_B/ag	4.000	< 0.20	< 0.20	None
87T	trans_hv/3ph	6.667	< 0.33	< 0.33	None
87T	trans_hv/ag	4.000	< 0.20	< 0.20	None
OC	bus_C_ld/3ph	4.348	1.304	1.304	None (gate inactive)
OC	bus_C_ld/ag	2.609	1.826	1.826	None (gate inactive)

The asymmetry arises because the line zone model has distributed-parameter effects (line capacitance, travelling wave) that are absent in the lumped bus and transformer models. At low current levels the modelling approximation degrades, reducing f_{int} .

6.2 Remediation Options

Three options exist to recover 87L HIF sensitivity:

1. **Magnitude-override extension to 87L.** Apply the same override rule used for 87B (TR-10): if $\hat{I}_{\text{diff, fund}} \geq I_{\text{ovr}}$, bypass the veto. However, the appropriate I_{ovr} for 87L must be below the fault current at $\alpha = 0.15$ (0.72 pu), suggesting $I_{\text{ovr}} \approx 0.30$ pu. This risks reduced selectivity against real CT saturation on through-faults.
2. **Improved 87L forward model.** Replace the lumped-parameter line model with a Π -section equivalent that includes line capacitance. This would improve model accuracy at low current levels and increase f_{int} for genuine internal faults.
3. **Dual-threshold gate.** Use a lower f_{int} threshold ($f_{\text{int, thresh}} = 0.40$ instead of 0.60) for the 87L zone, reducing the sensitivity penalty while accepting marginally reduced selectivity against mild CT saturation.

Option (2) is preferred as it addresses the root cause (model approximation quality) rather than the symptom (gate conservatism). It is recommended as the primary development item following this study.

6.3 Practical Significance

In a real power network, high-impedance faults with $I_f < 0.72$ pu are also below the practical detection limit of conventional OC protection (typically $I_{\text{op, min}} \approx 0.20$ pu differential, $I_{\text{OC, pickup}} \approx 0.5\text{--}1.0$ pu time-overcurrent). The HIF range where SAMBP has a sensitivity disadvantage relative to the conventional *differential* relay is $I_f \in [0.20, 0.72]$ pu (3.6–36% of nominal for line_mid/ag). Within this range, special HIF detection algorithms (spectral analysis, pattern recognition) would be required regardless of the protection scheme.

7 Conclusion

The HIF sensitivity margin study establishes quantitative lower bounds for SAMPB detection capability across all four protected zones:

- **87B, 87T:** TPR = 1.000 maintained to $\alpha = 0.05$ ($I_f \geq 0.20$ pu). No sensitivity penalty vs. conventional relay.
- **87L:** Stage-2 gate raises effective detection threshold from < 0.24 pu (conventional) to 0.56 pu (3ph) / 0.72 pu (ag). Root cause: low-SNR differential at reduced fault levels produces ambiguous LM solutions with $f_{\text{int}} < 0.60$.
- **OC:** Detection threshold determined analytically by the pickup threshold (1.5 pu RMS) and nominal fault current.

The 87L sensitivity penalty represents a quantified selectivity–sensitivity trade-off. Remediation via an improved distributed-parameter 87L forward model is recommended.

Availability. Script: `sampb/sampb_system/run_hif_study.py` (seed 2026).

References

- [1] SMBP TR-09/2026: Stage-2 gate recalibration and extended monte carlo validation. Technical report, PhD Thesis Supporting Report, 2026.
- [2] SMBP TR-10/2026: Ct saturation injection study. Technical report, PhD Thesis Supporting Report, 2026.